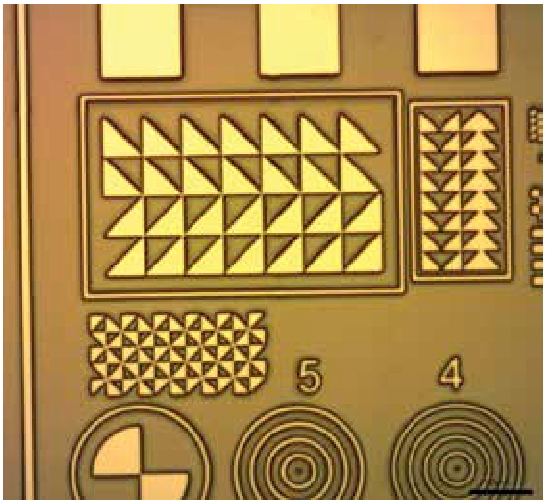
and other microscopic agents, and this method has worked for decades, but when looking for exceedingly smaller agents that are photosensitive, quantum dot lighting has proven to be far more accurate. Then there is the ability to inject quantum dots into tissue and monitor them for months on end, as the tissue grows and changes around them. They're even being considered for use as qubits in quantum computers, and are also being used to build much more efficient solar panels.

Biopolymers: A polymer is just a very large molecule, and when you tag on bio to the front of the term, it merely means polymers made by organic life. When it comes to the nano world, these biopolymers are being used in eco-friendly ways to manufacture nanomaterials, and are also being used as nanoparticles themselves. Think about it, if you could have nanoparticles or quantum dots injected into you to monitor a medical condition, would you rather they be inorganic plastics or organic biopolymers that your body can break down on its own? A no-brainer, right? But what if it's not just stuff being injected into your tissue, but your own tissue being used to "grow" a new organ? Nano biopolymers are definitely an exciting field that promises a lot for the future, and we will cover them in more detail later in this article.

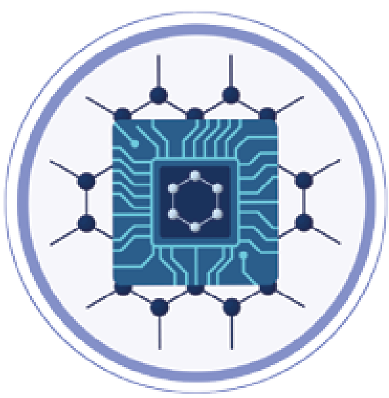
Fullerene: There's no denying how special carbon is to all life, given that we're all essentially the same carbon-based lifeforms. That said, carbon has quite a few more interesting qualities, especially when it comes to nanomaterials. For starters, carbon has plenty of allotropes, which means it has a pretty diverse set of materials that are all still just carbon at the end of the day. If chemistry isn't your strong point, an allotrope is different materials being formed based on how an element's atoms are arranged – this is why diamonds, graphite, and amorphous carbon (coal and soot) are all just carbon, but have such different properties. Fullerenes are named after Richard Buckminister Fuller (RBF), a famous American futurist, and are basically allotropes



Credit: Stanford University

A test wafer for photolithography. The thinnest line is about 1.5 um in thickness.

of carbon that have some exotic atomic bond shapes. The general shape is like that of a mesh, and these can be a flat mesh, a rolled up mesh, a tube-like structure, and many more. Of all fullerenes, C60 is the smallest molecule that forms a "ball", and was christened Buckminsterfullerene. This is because RBF had earlier imagined a geodesic dome or sphere – a sphere built by connecting polygons together, much like a football (soccer ball in some countries) is just patches of hexagons and pentagons stitched together to form an approximation of a sphere. C60 is shaped just like a football, and fullerenes that form such spherical or near-spherical "balls" are also known as Buckyballs (still named after RBF).



NANOMANUFACTURING

The creation of nanomaterial on a large scale is called nanomanufacturing, or nanofabrication. This includes all materials that have at least one, two or all three dimensions in the nano scale (1nm to 100nm). And it's not just about the creation of materials on their own, but also creating nanomaterial on top of, or inside regular objects also counts as nanomanufacturing. Obviously, the semiconductor industry (electronics, computers, gadgets, etc.) is the biggest manufacturer and consumer of nanomaterial, but it's pretty much all pervasive now.

There are two main ways in which nanomanufacturing is done, and we're going to dig deeper into them now.

Top down

Most manufacturing in ancient times was top down. Masons working with stone, or sculptors chiselling away at large blocks of marble to create masterpieces like the statue of David by Michelangelo. Nano manufacturing is often done in this way as well, using lasers to etch and chip away at larger structures to whittle down things to the nanoscale.

The first method of manufacturing is lithography, or in this case, nanolithography, and there are a few different methods used to accomplish this.

Optical: Also called photolithography, this method uses light of very short wavelengths (usually UV) to create patterns on ultra thin films of a substrate. It's like the manufacture of printed circuit boards, but at a much smaller size and with much more precision. In fact pretty much all of it is done by robots that have been specially built to do this.

The photolithography takes place in a dust, mote and organic (no bacteria or viruses even) free environment, and a wafer is thoroughly washed (with peroxide) to get rid of any possible contaminants. Then the wafer is heated to get rid of all cleaning liquids, to as high as 150